FROM GRID TO RINGS

Five cities, one founding centre each, and what is reported

Data	Rings	Results
Five cities Lagos 20 LGAs, Kano 16, Ibadan 11, Kaduna 4, Onitsha 5 (geoBoundaries)	**100 m rings** Each pixel split into 25 points, people counted by distance	**Where people live** Zone shares, 1990 to 2020. Section 6.1, Tables 1-2, Figure 3a
People GHS-POP 100 m, 1990 to 2020. WorldPop 2000 to 2020 as a check	**Remoteness** r = 1000 x km / sqrt(people 2020). Central r < 3, peri-urban r > 9.3	**City by city** Change plates and density profiles. Section 6.2, Figures 4-5
Centres A founding place for each city, from Iga Idunganran to Lugard Hall	**Spread factor** Q-Q slope of distance quantiles, 1% to 99%, 1990 against 2020	**Spread against expected** Observed Phi against the national fit. Section 6.3, Figure 3b
Yardsticks 183 GHSL cities urban by 1990; Mexico's 92, against its census	**Expected spread** ln Phi = beta ln growth + alpha dt, fitted on 183 cities, applied to five	**Three checks** Nigeria's 183 cities, Mexico, WorldPop. Sections 6.4-6.6, Figures 3c, 6

Every outline is a fixed group of local government areas, the same from 1990 to 2020.

Left: the sources. Middle: 100 m rings around each founding centre.
Right: where each strand is reported in the results.
The same steps run on 183 Nigerian cities and on Mexico, where a census says what happened.

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